HKU Restaurant Trip Time Estimation

COMP1110 Project Plan

2026/27 Semester 1 | 4 October 2026

1 Title and team members

Team members: Ali, Raul, Farid, Nirvana, Ly

2 Problem statement and solution concept

We propose a small program that helps HKU students choose where to eat by ranking restaurants according to the total time needed to have lunch and reach their destination classroom. The calculation includes travelling to the restaurant, queueing and ordering, food preparation, eating, and travelling to class. The initial proof of concept will use a small restaurant database and clearly labelled synthetic data to demonstrate the calculations and ranking.

Our design question is: How can a program combine travel and restaurant time estimates to rank lunch options by the earliest arrival at an HKU classroom? We will investigate how the ranking changes when queues, eating times, or destination classrooms change.

A student leaving class at 1:00 pm may want to know which lunch option gets them to their next classroom soonest. The closest restaurant may have a long queue, while a farther restaurant may serve food quickly. Comparing complete trips makes these trade-offs visible. A lecture deadline can be an optional secondary check; the main output is the restaurant ranking.

Proposed scope

We will include only a few restaurants to keep the project manageable, with no fixed restaurant count or visit-record target. A separate database will store HKU classroom entries from public sources, potentially covering all classrooms if information and time permit. We will begin with a small subset, without visiting rooms. Adding a classroom does not require adding another restaurant.

The user will select a starting location and destination classroom. The program will calculate each restaurant's total trip time and display restaurants in ascending order, with the shortest total first. Each result will show the synthetic travel and restaurant components. If departure time is supplied, it can also show a simulated arrival time. Location pairs without synthetic timing entries will be marked as unavailable.

The initial scope is a small proof of concept for restaurant ranking, not a campus-wide navigation service. We will use a simple route-time lookup for supported location pairs. A classroom directory may be broader than the location pairs included in our synthetic test scenarios. Automatic timetable integration, live queue monitoring, private restaurant order data, and a new mapping engine are outside the initial scope.

Intended outcome

We will produce a documented calculation and sorting method, small restaurant and classroom datasets, interface mock-ups, and worked test scenarios. A simple program will demonstrate the intended behaviour if implemented. We will explain the limits of synthetic inputs and how real measurements and wider coverage could develop the project further. A production-ready app is not required.

3 Computing and data science connection

This is a hybrid project combining system design, a simple ranking algorithm, and analysis of test scenarios. Structured data will link restaurants, classrooms, and route estimates. Arithmetic will combine the time components, and sorting will identify the shortest complete trips. Scenario analysis will show how changing an input affects the recommendation.

Restaurant classroom and synthetic travel data

The restaurant database will store identifiers, location references, and synthetic queueing, ordering, food preparation, and eating times. The classroom database will store room identifiers and buildings from public sources. A lookup table will link selected classroom and restaurant pairs with synthetic walking minutes in each direction. These values are test inputs, not measured travel times.

Stage	Time component
Outbound walk	Leaving the selected classroom to joining the restaurant queue.
Queue and ordering	Joining the queue to completing the order.
Food preparation	Completing the order to receiving the meal.
Eating and departure	Receiving the meal to leaving the restaurant, including seating or payment delays occurring here.
Return walk	Leaving the restaurant to reaching the next classroom.

All travel and restaurant timings in our tests will be synthetic, with hand-calculated expected totals. We will not conduct real-world route or restaurant timing tests. Location entries are database references only. We will check missing routes, duplicate identifiers, and negative times, and avoid double-counting the stages above.

Travel estimates and ranking method

For the initial demonstration, the program will retrieve synthetic outbound and onward walking times from a lookup table. Each direction will have its own test value, allowing different starting and destination classrooms. We will not inspect rooms, walk routes, or measure distances or travel times. A classroom entry will be usable in a test only when the required synthetic travel values are available.

Total trip time = travel to restaurant + queueing and ordering + food preparation + eating and departure + travel to destination classroom. The program will calculate this total for every restaurant with complete inputs, then sort the results from shortest to longest. Equal totals will be shown as ties, with a consistent secondary order such as restaurant name. For a common departure time, the smallest total gives the earliest estimated arrival at class.

For illustration only, Restaurant A could require 5 minutes outward travel, 25 minutes in the restaurant, and 10 minutes to class: 40 minutes overall. Restaurant B could require 8, 15, and 12 minutes respectively: 35 minutes overall. B ranks first despite its longer walking time. These are synthetic values, not observations.

Testing with synthetic scenarios

We will compare program outputs with hand-calculated totals and rankings for synthetic scenarios: long queues nearby, quick service farther away, tied totals, changed destinations, and missing or invalid data. Varying one input at a time will show when the preferred restaurant changes. Calculating totals requires work proportional to the number of restaurants; comparison sorting takes roughly n log n steps for n restaurants. These tests check functional correctness, not real-world prediction accuracy.

4 Literature and background review

Existing work covers restaurant choice, campus navigation, and crowdsourced queue estimation. The following five sources inform our design and define its context. Our initial demonstration will use synthetic timings, while the sources suggest ways to obtain and improve real estimates in a later version.

Restaurant visit information

Google describes popular times, live visit information, wait estimates, and typical visit duration based on aggregated, anonymized data from opted-in users [1]. Such information is available only when sufficient data exists. Wait definitions also vary: a sit-down restaurant's wait can mean time until seating. This makes visit information a useful comparison, but not a direct substitute for our consistently defined stages or a classroom-to-classroom time budget.

Campus location information

The HKU Campus Map lists buildings and facilities, including catering outlets and lifts [2]. It provides a public reference for identifying locations in the database. We will not use it to validate walking routes or travel times in this version. Synthetic travel values will let us test the ranking of complete lunch trips to a selected classroom.

Crowdsourced queue estimates

Goncalves and colleagues evaluated a kiosk-based crowdsourcing system in four university restaurants over 19 weekdays [3]. It collected queue estimates without interfacing with billing or customer systems and examined biases and report recency. This shows that crowdsourced restaurant waiting-time estimation is established. It suggests a possible future source of real inputs for our program. Our initial demonstration focuses on combining the whole lunch trip and ranking options for a specified classroom, without claiming to reproduce their field evaluation.

Automatic wait detection

LineKing combines smartphone-based wait detection with a cloud service and was deployed at a university coffee shop [4]. It demonstrates another existing approach to crowdsourced wait monitoring. Automatic sensing is outside our initial scope; synthetic wait values let us first test the ranking logic. The accuracy reported for that deployment does not establish the accuracy of our inputs.

Restaurant choice and travel

Athey and colleagues model lunchtime restaurant choices using mobile location data from several thousand anonymous users in the San Francisco Bay Area [5]. Their analysis incorporates restaurant preferences and willingness to travel. It supports considering travel alongside restaurant characteristics. Our simpler objective is to rank complete lunch trips by total time to a destination classroom, without estimating individual tastes or training a large statistical model.

Proposed research gap

Our proposed focus is a small, transparent comparison of complete HKU lunch trips: how do restaurant delays and classroom destinations change which option gets a student to class soonest? Existing queue and travel methods provide components for this design. The contribution we will investigate is their integration and explanation in a classroom-specific ranking, rather than a new queue predictor or sorting algorithm. We will use controlled scenarios to examine when proximity alone gives a different answer. This initial review does not establish that the combination has never been studied; the originality claim must remain modest and be refined through further comparison.

5 Feasibility limitations and ethics

The initial work needs a small structured dataset, a spreadsheet or simple program, and interface mock-ups. Synthetic timings remove the need for a large survey or access to restaurant order systems. A spreadsheet can independently check totals; a simple program can read the records, calculate each complete trip, and sort the results. We will limit the first version to methods the team can implement and explain within the course schedule.

Synthetic data is suitable for checking calculations and exploring scenarios, but cannot show that a student would actually take the predicted time. The small restaurant selection also prevents claims that the program finds the fastest lunch option across all of HKU. We will label all invented timings and report conclusions as outcomes under stated assumptions. A larger location directory cannot compensate for missing or inaccurate route-time data.

Classroom locations may be incomplete, and walking times vary with entrances, stairs, lifts, walking speed, and weather. We will begin with supported location pairs and state the coverage explicitly. The program will not guarantee on-time arrival or claim live queue awareness. Later development could add more restaurants and classrooms, timed walking routes, voluntary reports, time-of-day estimates, and comparisons against actual lunch trips. Those extensions are not prerequisites for the initial demonstration.

The initial synthetic tests will use no personal diner records or private restaurant order information. We will not invent customer identities or present synthetic values as restaurant performance evidence. Classroom and restaurant location information will be taken from suitable public sources and attributed. If voluntary reports are collected later, participants will be told the purpose, names and full timetables will not be needed, and results will be aggregated. Any collection involving people or restaurant premises will be planned with appropriate consent or permission before it begins.

6 Timeline and team roles

Timeline

Period in 2026	Milestone
4 October	Finalize the scope, synthetic-data approach, sources, and formatting; submit the Project Plan by 11:59 PM Hong Kong time.
5–11 October	Select a small restaurant set and classroom entries from public sources; define database fields and location pairs for synthetic tests.
12–19 October	Create labelled synthetic restaurant and travel times with expected totals; design the calculation and sorting logic and prepare mock-ups or a simple prototype.
20–24 October	Attend the interim progress meeting; demonstrate ranking with synthetic scenarios and explain the limits of the test data.
25 October–4 November	Run scenario tests, check totals and sorting, and examine how changed inputs affect the ranking; submit individual progress logs by 4 November.
5–15 November	Complete the report, test results, limitations, and future-development discussion; draft presentation slides. Submit the report and individual reflections by 15 November.
16–26 November	Finalize slides and rehearse the 10-minute presentation before the assigned slot on 17, 19, 24, or 26 November.

Team roles

Role	Responsibilities
Background research **Raul**	Review related systems and research, maintain references, and refine the contribution and scope.
Synthetic data and scenarios **Ali and Farid**	Create synthetic restaurant and travel times for a small dataset; document assumptions and define scenarios, including missing or invalid inputs.
Location database organization **Nirvana**	Organize restaurant and classroom entries from public sources and link them by identifiers. No site visits or route measurements.
Interface design **Ali**	Design mock-ups for selecting a starting location and destination classroom, and displaying restaurants ranked by total trip time with a breakdown of time components.
Calculation and testing **Ali and Raul**	Develop total-time and sorting logic; compare results with hand-calculated synthetic totals and test ties, changed destinations, and missing or invalid data.
Coordination and communication **Farid and Ly**	Track milestones, integrate and edit the report, document limitations and future development, and organize presentation preparation.

References

[1] Google. “About popular times, wait times & visit duration data.” Google Business Profile Help. https://support.google.com/business/answer/6263531?hl=en

[2] The University of Hong Kong. “HKU Campus Map.” https://www.maps.hku.hk/index.php?lang=en

[3] Goncalves, J., Kukka, H., Sánchez, I., and Kostakos, V. (2016). “Crowdsourcing Queue Estimations in Situ.” Proceedings of CSCW 2016, pp. 1040–1051. https://doi.org/10.1145/2818048.2819997

[4] Bulut, M., Yilmaz, Y., Demirbas, M., Ferhatosmanoglu, N., and Ferhatosmanoglu, H. (2013). “LineKing: Crowdsourced Line Wait-Time Estimation Using Smartphones.” Mobile Computing, Applications, and Services, MobiCASE 2012 Revised Selected Papers. https://doi.org/10.1007/978-3-642-36632-1_12

[5] Athey, S., Blei, D., Donnelly, R., Ruiz, F., and Schmidt, T. (2018). “Estimating Heterogeneous Consumer Preferences for Restaurants and Travel Time Using Mobile Location Data.” arXiv:1801.07826. https://arxiv.org/abs/1801.07826